



Fig. 2: Datasheet of VO2223A



radar sensing or solid-state relay applications.

This single compact design supports multiple modes of operation, allowing configuration for standalone use or full IoT integration.

and automatically activating them at night when motion is detected.

According to the optotriac data-sheet (refer Fig. 2), the triac can directly drive an AC bulb via pins 8 and 6. A snubber circuit may be added for inductive loads. Fig. 2 presents the VO2223A configuration circuit.

The design includes an FPC connector for IoT board integration and can be configured in four modes: Mode 1, Mode 2, Mode 3, and Mode 4.

Mode 1: Human radar SSR mode.

In this mode, the three-point solder jumper JP1 is shorted to trigger the optotriac based on human motion detected by the sensor. In this mode JP1 = 1 and 2 shorted and JP2 = 1 and 2 shorted.

In this standalone configuration, the circuit functions as a fully automatic motion-activated light switch. The RCWL-0516 radar sensor detects human motion using the Doppler effect, producing a high output at the OUT pin for about three seconds upon detection. This signal is routed directly to the optotriac (VO2223A-X007T) through solder jumper JP1, which shorts pins 1 and 3. As shown in Fig. 3, the green line runs from the radar OUT to JP1, then to $R1 = 120\Omega$ and optotriac pin 1. The optotriac's internal triac (pins 4-6) then drives the AC load, such as an LED or CFL bulb up to 1A RMS, via connector CON2.

A snubber network ($R3 = 100\Omega$ and $C3 = 100\text{nF}$ across pins 4-6) ensures stable operation, particularly for resistive loads. The CDS pin can be left open or connected to an LDR through a $10\text{k}\Omega$ divider for night-only operation. No FPC connection is required in this mode, making it suitable for basic home lighting automation. When JP1 is shorted, the circuit acts as a standalone motion-activated light switch. The radar sensor output drives the optotriac directly, turning on the connected bulb whenever motion is detected.

Mode 2: Human radar sensor

mode. In this mode, the board functions as a dedicated motion sensor and alert unit, bypassing the SSR section